ASSIGNMENT 14

Examine AndroidManifest.xml and Identify Risky Exported Components

1. Introduction

AndroidManifest.xml is a configuration file that describes an Android application's package information, permissions, and components, including Activities, Services, Broadcast Receivers, and Content Providers. A component marked as exported may be accessible to other applications, depending on Android platform rules and any permissions applied. Exported components are sometimes necessary, but unnecessary or insufficiently protected exposure can allow unauthorized actions or data access.

2. Objectives

- Understand the role of AndroidManifest.xml.
- Identify components declared as exported.
- Assess potential security risks from exposed components.
- Recommend secure configuration and access-control measures.

3. Requirements

- An APK or Android project that you own or are authorized to inspect
- Android Studio / APK Analyzer, JADX, or APKTool
- A computer for reviewing the manifest

Safety note: Perform this review only on applications you own or have explicit permission to assess. Do not attempt to access private data or invoke sensitive components in third-party applications.

4. Locating AndroidManifest.xml

For an Android Studio project, open app/src/main/AndroidManifest.xml. For an APK, open it with Android Studio APK Analyzer or decode it with APKTool:

```
apktool d app.apk -o decoded_app
```

Then open decoded_app/AndroidManifest.xml. Replace app.apk with the authorized APK path. The decoded manifest may differ from the original source manifest because build tools can merge and transform configuration.

5. Identifying Exported Components

Review each <activity>, <service>, <receiver>, and <provider> declaration. Look for android:exported, intent-filter declarations, and component-level permissions. The exported attribute determines whether other apps can access a component, subject to other Android security checks. Components with intent filters and older target SDK versions may have default behavior that differs

from modern Android requirements; inspect the effective manifest and target SDK rather than relying on assumptions.

```
<activity android:name=".AdminActivity" android:exported="true" />
```

The example above makes AdminActivity externally accessible. If this screen performs administrative actions and lacks authorization checks, the configuration may create a security risk. Exported status alone does not prove a vulnerability; the component's purpose and protections must be assessed.

6. Component-Specific Risk Analysis

Component	Potential Risk	What to Check
Activity	Sensitive screen may be launched by another app.	Does it enforce authentication and authorization?
Service	External app may trigger sensitive background work.	Is external access required and protected by permissions?
Broadcast Receiver	Untrusted broadcasts may trigger unintended behavior.	Is the receiver exported unnecessarily or missing sender restrictions?
Content Provider	Private data may be queried or modified.	Are read/write permissions and URI access properly restricted?

7. Security Assessment Procedure

1. Record the package name and target SDK information.
2. List each declared Activity, Service, Broadcast Receiver, and Content Provider.
3. Record the exported value and any required permissions.
4. Identify components that handle sensitive operations or private data.
5. Compare exposure with the intended application design.
6. Document evidence and classify only verified weaknesses as findings.

8. Security Recommendations

- Set android:exported="false" for components that do not need cross-application access.
- For components that must be exported, apply appropriate permissions and enforce authorization checks.
- Validate all incoming Intent extras, actions, URIs, and other external input.
- Restrict Content Provider access and avoid exposing sensitive records.
- Use explicit Intents for internal component communication where suitable.
- Retest the effective manifest and application behavior after making changes.

9. Results and Observation Table

Fill in this table using the manifest from the authorized application being reviewed. Do not assume a component is vulnerable only because it is exported.

Component Name	Exported?	Permission / Purpose	Risk and Recommendation
Activity	Record value	Record details	Assess actual risk
Service	Record value	Record details	Assess actual risk
Broadcast Receiver	Record value	Record details	Assess actual risk
Content Provider	Record value	Record details	Assess actual risk

10. Conclusion

Examining AndroidManifest.xml helps identify application components that may be accessible to other applications. Reviewing exported status, intent filters, permissions, component purpose, and authorization logic allows a tester to recognize potential risks such as unauthorized actions or data exposure. Unnecessary exposure should be disabled, while components that require external access should be protected with suitable permissions and input validation. The assessment must be based on the effective manifest and verified application behavior.

References

1. Android Developers - App manifest overview: https://developer.android.com/guide/topics/manifest/manifest-intro
2. Android Developers - Activity element: https://developer.android.com/guide/topics/manifest/activity-element
3. Android Developers - Security best practices: https://developer.android.com/privacy-and-security/security-best-practices